

Extra Unit C practice 1

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity C.

Formulas

• Integral formulas

- $\int cf(x)dx = c \int f(x)dx$
- $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
- $\int kdx = kx + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
- $\int \sin(x)dx = -\cos(x) + C$
- $\int \cos(x)dx = \sin(x) + C$
- $\int \sec^2(x)dx = \tan(x) + C$
- $\int \csc^2(x)dx = -\cot(x) + C$
- $\int \sec(x)\tan(x)dx = \sec(x) + C$
- $\int \csc(x)\cot(x)dx = -\csc(x) + C$
- $\int e^x dx = e^x + C$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int f(g(x))g'(x)dx = f(g(x)) + C$

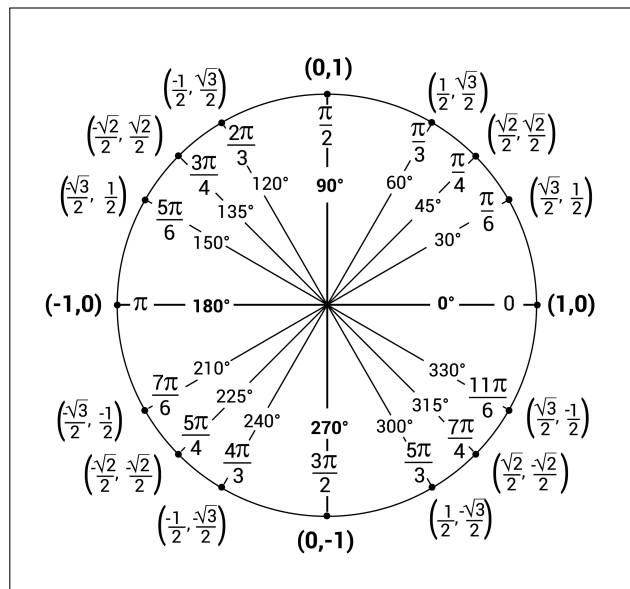
• $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

• Fundamental Theorem of Calculus

- $\int_a^b f(x)dx = F(b) - F(a)$
- $\frac{d}{dx} \left(\int_a^x f(t)dt \right) = f(x)$

• Law of natural growth/decay

- $Q'(t) = kQ(t)$
- $Q(t) = Q_0 e^{kt}$
- $T'(t) = k(T(t) - T_s)$



C1 - The integral

- (a) Consider the graph of a function $f(x)$ shown to the right.

- (i) Use a Riemann sum with $\Delta x = 3$ to find an overestimate of $\int_{-8}^1 f(x) dx$.

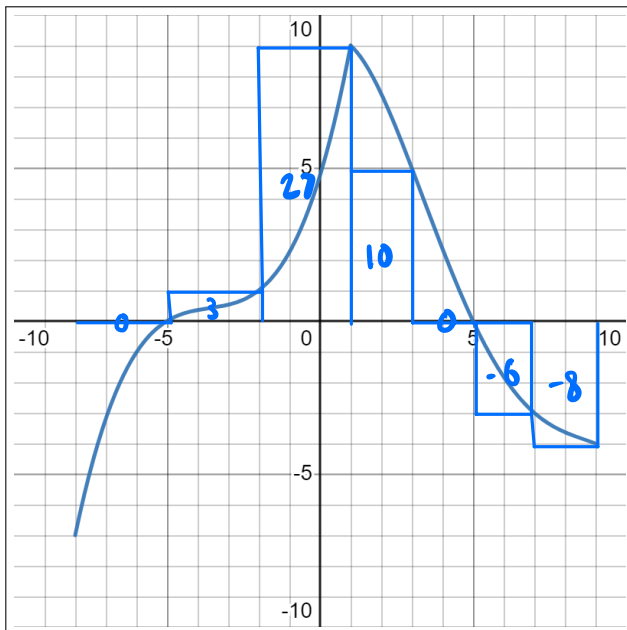
overestimate of inc. function: Right

$$0 + 3 + 27 = \boxed{30}$$

- (ii) Use a Riemann sum with 4 rectangles to find an underestimate of $\int_1^9 f(x) dx$

underestimate of dec. function: Right

$$10 + 0 - 6 - 8 = -4$$



- (b) Let $g(x)$ be the function

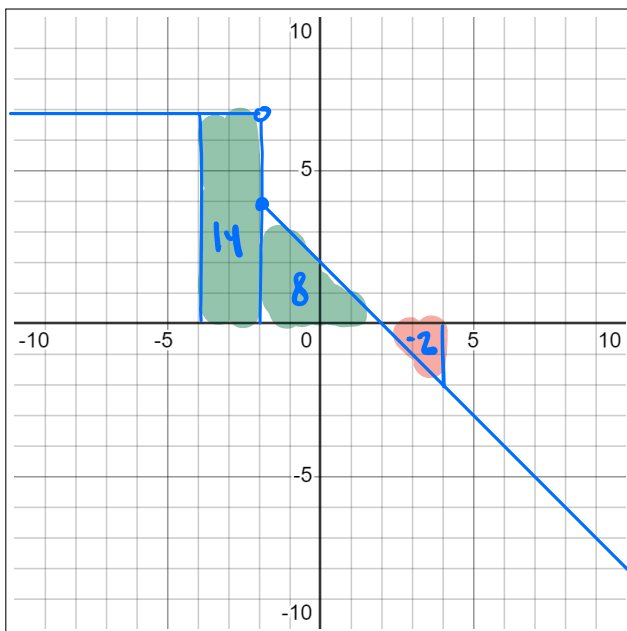
$$g(x) = \begin{cases} 7 & \text{if } x < -2 \\ -x + 2 & \text{if } x \geq -2 \end{cases}$$

Use area (not antiderivatives) to evaluate

$$\int_{-4}^4 g(x) dx$$

Show a rough sketch of the area(s) you used to answer the question on the axes to the right.

$$14 + 8 - 2 = \boxed{20}$$



C2 - The Fundamental Theorem of Calculus

(a) Find the derivative of the function:

$$f(x) = \int_{2x}^{3x+1} \sin(t^4) dt = \int_0^{3x+1} \sin(t^4) dt - \int_0^{2x} \sin(t^4) dt$$

$$f'(x) = \sin((3x+1)^4) \cdot 3 - \sin((2x)^4) \cdot 2$$

(b) Evaluate the following integral:

$$\int_0^1 (\sqrt[4]{u} + 1)^2 du$$

$$= \int_0^1 (\sqrt{u} + 2u^{\frac{1}{4}} + 1) du$$

$$= \frac{2}{3} u^{\frac{3}{2}} + \frac{8}{5} u^{\frac{5}{4}} + u \Big|_0^1$$

$$= \boxed{\frac{2}{3} + \frac{8}{5} + 1}$$

C3 - Integration by substitution

Find the following integrals. Clearly show each step of your work for full credit.

$$(a) \int_0^{\frac{\pi}{2}} \sin(x) \cos(x) e^{e^{\cos^2(x)} + \cos^2(x)} dx$$

$$u = \cos^2 x$$

$$u(0) = \cos^2(0) = 1$$

$$du = 2\cos(x) \cdot (-\sin x) dx$$

$$u\left(\frac{\pi}{2}\right) = \cos^2\left(\frac{\pi}{2}\right) = 0$$

$$dx = \frac{du}{-2\cos x \cdot \sin x}$$

$$\int_1^0 \sin(x) \cos(x) \cdot e^{e^u + u} \cdot \frac{du}{-2\cos x \sin x} = \frac{1}{2} \int_1^0 e^{e^u + u} du = \frac{1}{2} \int_1^0 e^{e^u} \cdot e^u du$$

$$w = e^u$$

$$w(1) = e$$

$$dw = e^u du$$

$$w(0) = 1$$

$$du = \frac{dw}{e^u}$$

$$\frac{1}{2} \int_e^1 e^w \cdot \frac{dw}{e^w} = \frac{1}{2} \int_e^1 e^w dw = \frac{1}{2} e^w \Big|_e^1$$

$$= \frac{1}{2} (e - e^e)$$

$$(b) \int \frac{x^3}{\sqrt{x^2 + 1}} dx$$

$$u = x^2 + 1 \rightarrow x^2 = u - 1$$

$$du = 2x dx$$

$$dx = \frac{du}{2x}$$

$$\int \frac{x^3}{\sqrt{u}} \cdot \frac{du}{2x} = \frac{1}{2} \int \frac{x^2}{\sqrt{u}} du = \frac{1}{2} \int \frac{u-1}{\sqrt{u}} du = \frac{1}{2} \int (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) du$$

$$= \frac{1}{2} \left(\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) + C$$

$$= \frac{1}{3} (x^2 + 1)^{\frac{3}{2}} - (x^2 + 1)^{\frac{1}{2}} + C$$

C4 - Inverse functions and their derivatives

(a) Let $f(x) = 2x + \sqrt{x^3 + 1}$

(i) Show that $f(x)$ is one-to-one on its domain.

$$x^3 + 1 \geq 0 \rightarrow x \geq -1$$

$$f'(x) = 2 + \frac{1}{2}(x^3 + 1)^{-\frac{1}{2}} \cdot 3x^2 \geq 2 \quad \text{for } x \geq -1$$

So $f(x)$ is always increasing, so it is one-to-one.

(ii) Find the derivative of $f^{-1}(x)$ at $x = 1$.

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))}$$

$$f(0) = 2 \cdot 0 + \sqrt{0^3 + 1} = 1 \quad \text{so } f^{-1}(1) = 0$$

$$= \frac{1}{f'(0)} = \frac{1}{2 + \frac{1}{2}(0^3 + 1)^{-\frac{1}{2}} \cdot 3 \cdot 0^2} = \boxed{\frac{1}{2}}$$

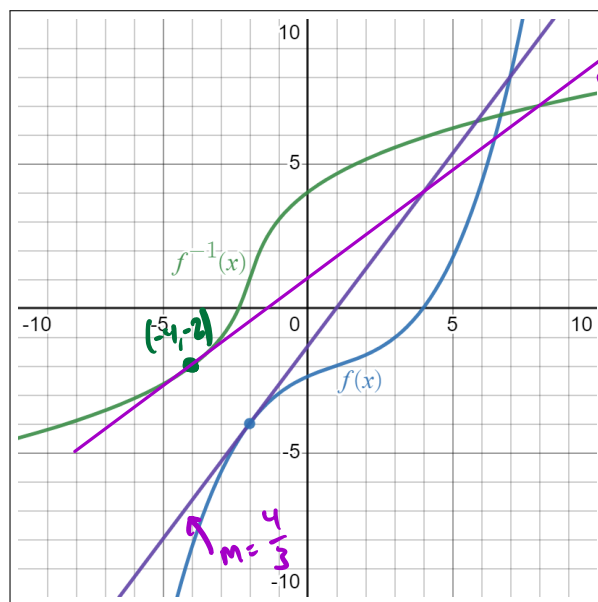
(b) The graphs of a function $f(x)$ and its inverse $f^{-1}(x)$ are shown below. The line tangent to $f(x)$ at $x = -2$ is also graphed.

(i) The tangent line shown can be used to determine the derivative of $f^{-1}(x)$ at some point. What is this point and what is the derivative? Your answer should be of the form

$$(f^{-1})'(a) = b$$

for some values of a and b .

$$(f^{-1})'(-4) = \frac{3}{4}$$



(ii) On the graph to the right, sketch the tangent line to $f^{-1}(x)$ at the point $x = a$ from your answer in part (i).

C5 - Exponential growth and decay

Cobalt-60 decays in such a way that at any moment the rate of change of its mass is -13% of its current mass per year. Suppose at time $t = 0$ years, a sample of Cobalt-60 has a mass of 100 milligrams.

- (a) What is the rate of change of this sample at $t = 0$?

$$Q'(t) = -0.13 Q(t)$$

$$Q'(0) = -0.13 \cdot Q(0) = -0.13 \cdot 100 = -13 \text{ mg/year}$$

- (b) How much mass does this sample lose in the first year of decay?

$$Q(t) = 100 e^{-0.13t}$$

$$Q(1) = 100 e^{-0.13 \cdot 1}$$

$$100 - 100 e^{-0.13}$$

- (c) What is the rate of change of this sample at $t = 1$?

$$Q'(1) = -0.13 \cdot Q(1)$$

$$= -0.13 \cdot 100 e^{-0.13}$$

$$= -13 e^{-0.13}$$